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Inventor(s)

Rotenberg; Alexis

System and Method of End-to-End Supply Chain Segmentation

Abstract

A system and method are disclosed for identifying supply chain segmentations from an initially non-segmented supply chain is disclosed. Embodiments of identifying supply chain segmentations includes receiving a current state of items in a supply chain network, wherein an inventory of the one or more supply chain entities is used to store one or more items, organizing one or more supply chain entities into one or more customer-product clusters, associating supply chain strategies for customer-product clusters, and transporting items among one or more supply chain entities.

Inventors: Rotenberg; Alexis (Egham, GB)

Applicant: Blue Yonder Group, Inc. (Scottsdale, AZ)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 15/724,108, filed Oct. 3, 2017, entitled “System and Method of End-to-End Supply Chain Segmentation,” which is a continuation-in-part of U.S. patent application Ser. No. 15/724,052, filed on Oct. 3, 2017, entitled “System and Method of End-to-End Supply Chain Segmentation,” which is a continuation-in-part of U.S. patent application Ser. No. 15/723,554, filed on Oct. 3, 2017, entitled “System and Method of End-to-End Supply Chain Segmentation,” now U.S. Pat. No. 12,293,316, which claims the benefit under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/403,576, filed Oct. 3, 2016, entitled “System and Method of End-to-End Supply Chain Segmentation.” U.S. patents application Ser. Nos. 15/724,108, 15/724,052, U.S. Pat. No. 12,293,316, and U.S. Provisional Application No. 62/403,576 are assigned to the assignee of the present application.

TECHNICAL FIELD

[0002] The present disclosure relates generally to supply chain segmentation and specifically to a system and method of segmenting a supply chain network based on customer and product attributes.

BACKGROUND

[0003] Many companies struggle to differentiate service to customers for their products in a consistent way. Standard supply chain processes, combined with a multitude of customers and products, make it difficult to deliver special services without workarounds in order-taking and planning systems. Determining how deep and granular service differentiation should be is difficult because having too much leads to complexity, while having too little does not differentiate a service offering sufficiently. These drawbacks are undesirable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A more complete understanding of the present invention may be derived by referring to the detailed description when considered in connection with the following illustrative figures. In the figures, like reference numbers refer to like elements or acts throughout the figures.

[0005] FIG. 1 illustrates an exemplary supply chain network, according to a first embodiment;

[0006] FIG. 2 illustrates the segmentation analyzer of FIG. 1 in greater detail, according to an embodiment;

[0007] FIG. 3 illustrates an exemplary segmentation strategies method of the segmentation strategies module of the segmentation analyzer of FIG. 2, according to an embodiment;

[0008] FIG. 4 illustrates a simplified segmented supply chain network with various exemplary supply chain models and various exemplary steering options, according to an embodiment;

[0009] FIG. 5 illustrates a differentiator hierarchy for module configuration, according to an embodiment;

[0010] FIG. 6 illustrates an exemplary overview of supply chain planning and execution processes with various exemplary supply chain strategies and levers, according to an embodiment;

[0011] FIG. 7 illustrates an exemplary customer hierarchy, according to an embodiment;

[0012] FIG. 8 illustrates an exemplary allocation logic for various supply chain models associated with customer hierarchy, according to an embodiment;

[0013] FIG. **9** illustrates supply chain strategies selected to differentiate supply chain models, according to an embodiment;

[0014] FIG. **10** illustrates supply chain levers associated with various supply chain strategies and configuration options, according to an embodiment;

[0015] FIGS. **11A** and **11B** illustrate particular configuration options for various exemplary supply chain models, according to an embodiment;

[0016] FIG. **12** illustrates configuration options for levers for a strategic supply chain model, according to an embodiment;

[0017] FIG. **13** illustrates configuration options for levers for a collaborative supply chain model, according to an embodiment;

[0018] FIG. **14** illustrates configuration options for levers for an agile supply chain model, according to an embodiment;

[0019] FIG. **15** illustrates configuration options for levers of a responsive supply chain model, according to an embodiment;

[0020] FIG. **16** illustrates configuration options for levers for an efficient supply chain model, according to an embodiment; and

[0021] FIG. **17** illustrates a chart detailing the relationship between supply chain strategies, levers, and supply chain strategy steering.

DETAILED DESCRIPTION

[0022] Aspects and applications of the invention presented herein are described below in the drawings and detailed description of the invention. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts.

[0023] In the following description, and for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the various aspects of the invention. It will be understood, however, by those skilled in the relevant arts, that the present invention may be practiced without these specific details. In other instances, known structures and devices are shown or discussed more generally in order to avoid obscuring the invention. In many cases, a description of the operation is sufficient to enable one to implement the various forms of the invention, particularly when the operation is to be implemented in software. It should be noted that there are many different and alternative configurations, devices and technologies to which the disclosed inventions may be applied. The full scope of the inventions is not limited to the examples that are described below.

[0024] One of the reasons companies struggle with creating differentiated services is because service offerings are a multi-dimensional problem dependent on competing priorities. Customers have different service priorities (e.g. higher fill rates at a premium price, extra services, stripped-down services, and the like) while also having different cost priorities (e.g. service package cost and the like). These challenges make it difficult to design a supply chain that offers consistent service to groups of customers.

[0025] Supply chain segmentation may aid companies to meet the needs of customers by customizing services to the customers' attributes, which often vary greatly. According to some embodiments, segmentation may be implemented in the logistics-phase of supply chain management by providing different package sizes and package assortments to customer with different shelf requirements. For example, a beauty care products distributor could send a large box of a single type of shampoo to a big-box retailer but send a smaller box containing an assortment of beauty care products to a mom-and-pop store with limited shelf and storage space. According to a further example, a vehicle manufacturer may set up additional production lines to meet the needs of customers that desire customization of the vehicle without slowing down a production line that produces fully-standard vehicles. For example, a vehicle manufacturer could create three production lines that run parallel to each other: a first production line for fully-customized vehicles,

a second production line for slightly-customized vehicles, and a third production line for fully-standard vehicles. In this way, the service requirements of each customers' needs may be met without significant impact to other customers.

[0026] These segmentation techniques do not, however, indicate how to segment customers, how many segments to create, the characteristics of each segment, or how to implement the segments into a supply chain. In addition, these supply chain segmentation techniques use technical parameters, such as demand variability or profitability, but cannot tie back to customer service requirements, nor validate why the segmentation was based on certain parameters and not on others. The supply chain models created by these segmentation techniques are loosely defined and derived typically from product characteristics. No consistent approach exists to segment a supply chain and define the supply chain models' relationship to customer service objectives.

[0027] According to embodiments, embodiments of the current disclosure comprise resolving priority conflicts across a segmented supply chain, defining clusters of customers and products, and determining particular implementations of demand planning, supply and distribution planning, inventory management, allocation planning, and order fulfilment. According to embodiments, customers and products may be classified to directly configure service priorities in segmented supply chain management planning and execution modules.

[0028] According to a further aspect, embodiments of the current disclosure comprise mapping customer clusters, product clusters, customer-product clusters, and/or customer business models to supply chain models. These mappings are complicated by the fact that all customers may potentially order all products and not all customers may want the same service for all products that they order. Additionally, the particular supply chain function that is being performed may shift the priority from customer to product, or vice versa. For example, the closer the supply chain functions are to the market side of the supply chain (such as logistics), the more likely a customer dimension will be prioritized. However, as the supply chain functions are closer to the factories and the suppliers of raw material, the more likely a product dimension will be prioritized.

[0029] FIG. 1 illustrates exemplary supply chain network **100** according to a first embodiment. Supply chain network **100** comprises segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, computer **160**, network **170**, and communication links **180-190**. Although a single segmentation analyzer **110**, a single inventory system **120**, one or more imagers **130**, a single transportation network **140**, one or more supply chain entities **150**, a single computer **160**, and a single network **170**, are shown and described, embodiments contemplate any number of segmentation analyzers, inventory systems, imagers, transportation systems, supply chain entities, computers, or networks, according to particular needs.

[0030] In one embodiment, segmentation analyzer **110** comprises server **112** and database **114**. According to embodiments, server **112** comprises one or more modules for segmenting supply chains according to various approaches. For example, and as discussed in more detail below, the one or more modules define supply chain models based, at least in part, on customer business models, KPI trade-offs, and supply chain characteristics. In addition, these one or more modules define clusters of customers and products and determines particular implementations of demand planning, supply and distribution planning, inventory management, allocation planning, and order fulfilment. Furthermore, the one or more modules configure supply chain processes with levers, enablers, and configuration options to enforce one or more supply chain models to customers, products, or customer-product clusters. Each of these modules will be discussed in more detail below.

[0031] Inventory system **120** comprises server **122** and database **124**. Server **122** of inventory system **120** is configured to receive and transmit item data, including item identifiers, pricing data, attribute data, inventory levels, and other like data about one or more items at one or more locations in the supply chain network **100**. Server **122** stores and retrieves item data from database

124 or from one or more locations in supply chain network **100**.

[0032] One or more imagers **130** comprise an electronic device that receives imaging information from one or more sensors **136** or from one or more databases in supply chain network **100**.

According to embodiments, one or more imagers **130** comprise one or more processors **132**, memory **134**, one or more sensors **136**, and may include any suitable input device, output device, fixed or removable computer-readable storage media, or the like. According to embodiments, one or more imagers **130** identify items near one or more sensors **136** and generate a mapping of the item in supply chain network **100**. As explained in more detail below, inventory system **120** and transportation network **140** use the mapping of an item to locate the item in supply chain network **100**. The location of the item is then used to coordinate the storage and transportation of items in supply chain network **100** to implement one or more prioritizations, hierarchies, and/or supply chain models generated by segmentation analyzer **110**.

[0033] One or more imagers **130** may comprise a mobile handheld device such as, for example, a smartphone, a tablet computer, a wireless device, or the like. In addition, or as an alternative, one or more imagers **130** comprise one or more networked electronic devices configured to transmit item identity information to one or more databases as an item passes by or is scanned by one or more imagers **130**. This may include, for example, a stationary scanner located at one or more supply chain entities **150** that identifies items as the items pass near the scanner. One or more sensors **136** of one or more imagers **130** may comprise an imaging sensor, such as, a camera, scanner, electronic eye, photodiode, charged coupled device (CCD), or any other electronic or manual sensor that detects images of objects. In addition, or as an alternative, one or more sensors **136** may comprise a radio receiver and/or transmitter configured to read an electronic tag, such as, for example, a radio-frequency identification (RFID) tag. In addition, or as an alternative, each of the one or more items may be represented in supply chain network **100** by an identifier, including, for example, Stock-Keeping Unit (SKU), Universal Product Code (UPC), serial number, barcode, tag, RFID, or any other object that encodes identifying information. As discussed above, one or more imagers **130** may generate a mapping of one or more items in the supply chain network **100** by scanning an identifier or object associated with an item and identifying the item based, at least in part, on the scan.

[0034] Transportation network **140** comprises server **142** and database **144**. According to embodiments, transportation network **140** directs one or more transportation vehicles **146** to ship one or more items between one or more supply chain entities **150**, based, at least in part, on the one or more prioritizations, hierarchies, and/or supply chain models generated by segmentation analyzer **110**. Transportation vehicles **146** comprise, for example, any number of trucks, cars, vans, boats, airplanes, unmanned aerial vehicles (UAVs), cranes, robotic machinery, or the like.

Transportation vehicles **146** may comprise radio, satellite, or other communication that communicates location information (such as, for example, geographic coordinates, distance from a location, global positioning satellite (GPS) information, or the like) with segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, and/or one or more supply chain entities **150** to identify the location of the transportation vehicle **146** and the location of any inventory or shipment located on the transportation vehicle **146**. In addition to the supply chain models, the number of items shipped by transportation vehicles **146** in transportation network **140** may also be based, at least in part, on the number of items currently in stock at one or more supply chain entities **150**, the number of items currently in transit in the transportation network **140**, forecasted demand, a supply chain disruption, or the like.

[0035] As shown in FIG. 1, supply chain network **100** operates on one or more computers **160** that are integral to or separate from the hardware and/or software that support segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, and one or more supply chain entities **150**. Supply chain network **100** comprising segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, and one or more

supply chain entities **150** may operate on one or more computers that are integral to or separate from the hardware and/or software that support segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, and one or more supply chain entities **150**. Computers **160** may include any suitable input device **162**, such as a keypad, mouse, touch screen, microphone, or other device to input information. Output device **164** may convey information associated with the operation of supply chain network **100**, including digital or analog data, visual information, or audio information. Computer **160** may include fixed or removable computer-readable storage media, including a non-transitory computer readable medium, magnetic computer disks, flash drives, CD-ROM, in-memory device or other suitable media to receive output from and provide input to supply chain network **100**.

[0036] Computer **160** may include one or more processors **166** and associated memory to execute instructions and manipulate information according to the operation of supply chain network **100** and any of the methods described herein. In addition, or as an alternative, embodiments contemplate executing the instructions on computer **160** that cause computer **160** to perform functions of the method. Further examples may also include articles of manufacture including tangible computer-readable media that have computer-readable instructions encoded thereon, and the instructions may comprise instructions to perform functions of the methods described herein. According to some embodiments, the functions and methods described in connection with one or more imagers **130** may be emulated by one or more modules configured to perform the functions and methods as described.

[0037] In addition, and as discussed herein, supply chain network **100** may comprise a cloud-based computing system having processing and storage devices at one or more locations, local to, or remote from segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, and one or more supply chain entities **150**. In addition, each of the one or more computers **140** may be a work station, personal computer (PC), network computer, notebook computer, tablet, personal digital assistant (PDA), cell phone, telephone, smartphone, wireless data port, augmented or virtual reality headset, or any other suitable computing device. In an embodiment, one or more users may be associated with the inventory planer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, and one or more supply chain entities **150**.

[0038] These one or more users may include, for example, a “manager” or a “planner” handling supply chain segmentation and/or one or more related tasks within the system. In addition, or as an alternative, these one or more users within the system may include, for example, one or more computers programmed to autonomously handle, among other things, one or more supply chain processes such as demand planning, supply and distribution planning, inventory management, allocation planning, order fulfilment, and adjustment of manufacturing and inventory levels at various stocking points and distribution centers, and/or one or more related tasks within supply chain network **100**.

[0039] One or more supply chain entities **150** represent one or more supply chain networks, including one or more enterprises, such as, for example networks of one or more suppliers **152**, manufacturers **154**, distribution centers **156**, retailers **158** (including brick and mortar and online stores), customers, and/or the like. Suppliers **152** may be any suitable entity that offers to sell or otherwise provides one or more items (i.e., materials, components, or products) to one or more manufacturers **154**. Suppliers **152** may comprise automated distribution systems **153** that automatically transport products to one or more manufacturers **154** based, at least in part, on one or more prioritizations, hierarchies, and/or supply chain models determined by segmentation analyzer **110** and/or one or more other factors described herein.

[0040] Manufacturers **154** may be any suitable entity that manufactures at least one product. Manufacturers **154** may use one or more items during the manufacturing process to produce any manufactured, fabricated, assembled, or otherwise processed item, material, component, good, or

product. In one embodiment, a product represents an item ready to be supplied to, for example, one or more supply chain entities **150** in supply chain network **100**, such as retailers **158**, an item that needs further processing, or any other item. Manufacturers **154** may, for example, produce and sell a product to suppliers **152**, other manufacturers **154**, distribution centers **156**, retailers **158**, a customer, or any other suitable person or entity. Manufacturers **154** may comprise automated robotic production machinery **155** that produce products based, at least in part, on one or more prioritizations, hierarchies, and/or supply chain models determined by segmentation analyzer **110** and/or one or more other factors described herein.

[0041] Distribution centers **156** may be any suitable entity that offers to store or otherwise distribute at least one product to one or more retailers **158** and/or customers. Distribution centers **156** may, for example, receive a product from a first one or more supply chain entities **150** in supply chain network **100** and store and transport the product for a second one or more supply chain entities **150**. Distribution centers **156** may comprise automated warehousing systems **157** that automatically remove products from and place products into inventory based, at least in part, one or more prioritizations, hierarchies, and/or supply chain models determined by segmentation analyzer **110** and/or one or more other factors described herein.

[0042] Retailers **158** may be any suitable entity that obtains one or more products to sell to one or more customers. Retailers **158** may comprise any online or brick-and-mortar store, including stores with shelving systems **159**. Shelving systems may comprise, for example, various racks, fixtures, brackets, notches, grooves, slots, or other attachment devices for fixing shelves in various configurations. These configurations may comprise shelving with adjustable lengths, heights, and other arrangements, which may be adjusted by an employee of retailers **158** based on computer-generated instructions or automatically by machinery to place products in a desired location in retailers **158**.

[0043] Although one or more supply chain entities **150** are shown and described as separate and distinct entities, the same entity may simultaneously act as any one of the one or more supply chain entities **150**. For example, one or more supply chain entities **150** acting as a manufacturer can produce a product, and the same one or more supply chain entities **150** can act as a supplier to supply an item to itself or another one or more supply chain entities **150**. Although one example of a supply chain network **100** is shown and described, embodiments contemplate any configuration of supply chain network **100**, without departing from the scope described herein.

[0044] In one embodiment, segmentation analyzer **110** may be coupled with network **170** using communications link **180**, which may be any wireline, wireless, or other link suitable to support data communications between segmentation analyzer **110** and network **170** during operation of supply chain network **100**. Inventory system **120** may be coupled with network **170** using communications link **182**, which may be any wireline, wireless, or other link suitable to support data communications between inventory system **120** and network **170** during operation of supply chain network **100**. One or more imagers **130** are coupled with network **170** using communications link **184**, which may be any wireline, wireless, or other link suitable to support data communications between one or more imagers **130** and network **170** during operation of distributed supply chain network **100**. Transportation network **140** may be coupled with network **170** using communications link **186**, which may be any wireline, wireless, or other link suitable to support data communications between transportation network **140** and network **170** during operation of supply chain network **100**. One or more supply chain entities **150** may be coupled with network **170** using communications link **188**, which may be any wireline, wireless, or other link suitable to support data communications between one or more supply chain entities **150** and network **170** during operation of supply chain network **100**. Computer **160** may be coupled with network **170** using communications link **190**, which may be any wireline, wireless, or other link suitable to support data communications between computer **160** and network **170** during operation of supply chain network **100**.

[0045] Although communication links **180-190** are shown as generally coupling one or segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and computer **160** to network **170**, any of segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and computer **160** may communicate directly with each other, according to particular needs.

[0046] In another embodiment, network **170** includes the Internet and any appropriate local area networks (LANs), metropolitan area networks (MANs), or wide area networks (WANs) coupling segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and computer **160**. For example, data may be maintained locally to, or externally of, segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and computer **160** and made available to one or more associated users of segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and computer **160** using network **170** or in any other appropriate manner. For example, data may be maintained in a cloud database at one or more locations external to segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and computer **160** and made available to one or more associated users of segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and computer **160** using the cloud or in any other appropriate manner. Those skilled in the art will recognize that the complete structure and operation of network **170** and other components within supply chain network **100** are not depicted or described. Embodiments may be employed in conjunction with known communications networks and other components.

[0047] In accordance with the principles of embodiments described herein, segmentation analyzer **110** may generate one or more prioritizations, hierarchies, and/or supply chain models that determine the routing, storage, and handling for the inventory of one or more supply chain entities **150** in supply chain network **100**. Additionally, segmentation analyzer **110**, inventory system **120**, imagers **130**, transportation network **140**, and/or one or more supply chain entities **150** may comprise a server and database comprising one or more modules to perform one or more planning and execution processes including demand planning processes, supply planning processes, demand fulfillment processes, an order entry system, comprising an order entry process, an allocation-planning order promising (AP-OP) engine, and the like, as described in more detail below.

Furthermore, segmentation analyzer **110**, inventory system **120**, and/or transportation network **140** may instruct automated machinery (i.e., robotic warehouse systems, robotic inventory systems, automated guided vehicles, mobile racking units, automated robotic production machinery, robotic devices and the like) to adjust product mix ratios, inventory levels at various stocking points, production of products of manufacturing equipment, proportional or alternative sourcing of one or more supply chain entities **150**, and the configuration and quantity of packaging and shipping of products based on one or more prioritizations, hierarchies, and/or supply chain models generated by segmentation analyzer **110**, plans and policies and/or current inventory or production levels.

[0048] For example, the methods described herein may include computers receiving product data from automated machinery having at least one sensor and the product data corresponding to an item detected by one or more imagers **130** of the automated machinery. The received product data may include an image of the item, an identifier, as described above, and/or other product data associated with the item (dimensions, texture, estimated weight, and any other like data). The method may further include computers looking up the received product data in a database system associated with segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, and/or transportation network **140** to identify the item corresponding to the product data received from the automated machinery.

[0049] The computers may also receive, from the automated machinery, a current location of the identified item. Based on the identification of the item, computers may also identify (or alternatively generate) a first mapping in the database system, where the first mapping is associated with the current location of the item. Computers may also identify a second mapping in the database system, where the second mapping is associated with a past location of the identified item. Computers may also compare the first mapping and the second mapping to determine if the current location of the identified item in the first mapping is different than the past location of the identified item in the second mapping. Computers may then send instructions to the automated machinery based, at least in part, on one or more differences between the first mapping and the second mapping such as, for example, to locate item to add to or remove from an inventory of or shipment for one or more supply chain entities **150**. In addition, or as an alternative, segmentation analyzer **110** monitors KPIs at one or more supply chain entities **150** and adjusts the orders and/or inventory of the one or more supply chain entities **150** based on the KPIs, one or more prioritizations, hierarchies, and/or supply chain models.

[0050] FIG. 2 illustrates segmentation analyzer **110** of FIG. 1 in greater detail, according to an embodiment. As discussed above, segmentation analyzer **110** may comprise one or more computers **160** at one or more locations including associated input devices **162**, output devices **164**, non-transitory computer-readable storage media, processors **166**, memory, or other components for receiving, processing, storing, and communicating information according to the operation of supply chain network **100**. Additionally, segmentation analyzer **110** comprises server **112** and database **114**. Although segmentation analyzer **110** is shown as comprising a single computer **160**, a single server **112** and a single database **114**, embodiments contemplate any suitable number of computers, servers, or databases internal to or externally coupled with segmentation analyzer **110**. According to some embodiments, segmentation analyzer **110** may be located internal to one or more retailers **158** of one or more supply chain entities **150**. In other embodiments, segmentation analyzer **110** may be located external to one or more retailers **158** of one or more supply chain entities **150** and may be located in for example, a corporate headquarters of the one or more retailers **158**, according to particular needs.

[0051] Server **112** of segmentation analyzer **110** may comprise value offering module **200**, clustering module **202**, and segmentation strategies module **204**. Value offering module **200** may define supply chain models based, at least in part, on customer business models, KPI trade-offs, and supply chain characteristics. According to embodiments, value offering module **200** identifies the number and type of supply chain models with specific service packages that meet customer and product priorities and which are aligned with the customers' and company's objectives. According to these embodiments, value offering module **200** avoids overly complex customizations of other segmentation approaches by generating a particular number of supply channels with the same number of specific customizations that meet all the needs of the company, instead of a one-size-fits-all solution. According to some embodiments, each of the customizations include particular configurations of supply chain processes or user interfaces that support the supply channel.

[0052] Embodiments of clustering module **202** define customer clusters, product clusters, and customer-product combination clusters (customer-product clusters), map customer clusters, product clusters, customer-product clusters, and/or customer business models to supply chain models, and automatically configure particular implementations of demand planning, supply and distribution planning, inventory management, allocation planning, and order fulfillment, including resolving priority conflicts across a segmented supply chain. According to embodiments, clustering module **202** classifies customers and products, combining Pareto analysis and leveraging different dimensions, to directly reflect different service priorities in the design and configuration of segmented supply chain planning and execution modules. Clustering module **202** may embed the cluster definitions in the segmented supply chain planning and execution modules by a customer view of a user interface.

[0053] Segmentation strategies module **204** configures supply chain processes with levers, enablers, and configuration options to enforce one or more supply chain models to customers, products, or customer-product clusters, as described in more detail below.

[0054] Although server **112** is shown and described as comprising a single value offering module **200**, a single clustering module **202**, and a single segmentation strategies module **204**, embodiments contemplate any suitable number or combination of these located at one or more locations, local to, or remote from segmentation analyzer **110**, such as on multiple servers or computers at any location in supply chain network **100**.

[0055] Database **114** of segmentation analyzer **110** may comprise one or more databases or other data storage arrangement at one or more locations, local to, or remote from, server **112**. Database **114** comprises, for example, inventory data **210**, supply chain data **212**, KPI library data **214**, inventory policies **216**, customer business models **218**, supply chain models **220**, and target service levels **222**. Although, database **114** is shown and described as comprising inventory data **210**, supply chain data **212**, KPI library data **214**, inventory policies **216**, customer business models **218**, supply chain models **220**, and target service levels **222**, embodiments contemplate any suitable number or combination of these, located at one or more locations, local to, or remote from, segmentation analyzer **110** according to particular needs.

[0056] Inventory data **210** of database **114** may comprise any data relating to current or projected inventory quantities or states. For example, inventory data **210** may comprise the current level of inventory for each item at one or more stocking points across supply chain network **100**. In addition, inventory data **210** may comprise order rules that describe one or more rules or limits on setting an inventory policy, including, but not limited to, a minimum order quantity, a maximum order quantity, a discount, and a step-size order quantity, and batch quantity rules. According to some embodiments, segmentation analyzer **110** accesses and stores inventory data **210** in database **114**, which may be used by segmentation analyzer **110** to place orders, set inventory levels at one or more stocking points, initiate manufacturing of one or more components, or the like. In addition, or as an alternative, inventory data **210** may be updated by receiving current item quantities, mappings, or locations from inventory system **120**, one or more imagers **130**, and/or transportation system **140**.

[0057] Supply chain data **212** may comprise any data of the one or more supply chain entities **150** including, for example, item data, identifiers, metadata (comprising dimensions, hierarchies, levels, members, attributes, cluster information, and member attribute values), fact data (comprising measure values for combinations of members), business constraints, goals and objectives of one or more supply chain entities **150**. KPI library **214** comprises one or more KPIs that are currently measured by the company and/or KPIs that may be measured by the company with existing IT solutions with consideration given to the objectives of any ongoing business transformation projects. According to embodiments, KPIs in KPI library **214** are identified from a standard reference framework for supply chain processes such as those from, for example, the Supply Chain Operational Reference Model (SCOR), American Production and Inventory Control Society (APICS), American Quality and Productivity Centre (AQPC), or the Gartner KPI framework. Although particular frameworks have been identified, embodiments contemplate any suitable framework or standards, according to particular needs.

[0058] Inventory policies **216** of database **114** may comprise any suitable inventory policy describing the reorder point and target quantity, or other inventory policy parameters that set rules for segmentation analyzer **110** to manage and reorder inventory based, at least in part, on customer business models, KPI trade-offs, and supply chain characteristics. According to embodiments, inventory policies **216** may be used by segmentation analyzer **110** to determine a no-stockout probability, fill rate, cost, or other like determination, as described below.

[0059] Customer business models **218** comprise groups of customers, channels, and/or product concepts based on one or more customer attributes. As explained in more detail below, customer

business models may comprise groups which share a particular set of customer attributes. These customer attributes may include, for example, ordering behavior, order pattern, leadtime expectations, product or packaging adjustments, and unique or particular service requests. More particular examples of customer attributes include: having limited or no storage space, requesting one or more suppliers **152** to keep consignment inventory, selling products from a competitor, purchasing items ahead of season, and placing large orders.

[0060] Supply chain models **220** comprise characteristics of a supply chain setup to deliver the customer expectations of a particular customer business model. These characteristics may comprise differentiating factors, such as, for example, MTO (Make-to-Order), ETO (Engineer-to-Order) or MTS (Make-to-Stock). However, supply chain models **220** may also comprise characteristics that specify the supply chain structure in even more detail, including, for example, specifying the type of collaboration with the customer (e.g. Vendor-Managed Inventory (VMI)), from where products may be sourced, and how products may be allocated, shipped, or paid for, by particular customers. Each of these characteristics may lead to a different supply chain model.

[0061] Target service level data **222** of database **114** may be calculated by segmentation analyzer **110** from a forecasted demand to ensure that a service level of one or more supply chain entities **150** is met with a certain probability. Target service level data **222** may comprise any suitable service level target. According to some embodiments, one or more supply chain entities **150** set a service level at 95%, meaning the one or more supply chain entities **150** will set the desired inventory stock level at a level that meets demand 95% of the time. Although, a particular service level target and percentage is described; embodiments contemplate any service target or level, for example, a service level of approximately 99% through 90%, a 75% service level, or any suitable service level, according to particular needs. Other types of service levels associated with inventory quantity or order quantity may comprise, but are not limited to, a maximum expected backlog and a fulfillment level. Once the service level is set, segmentation analyzer **110** may determine a replenishment order according to one or more replenishment rules, which, among other things, indicates to one or more supply chain entities **150** to determine or receive inventory to replace the depleted inventory.

[0062] FIG. 3 illustrates exemplary segmentation strategies method **300** of segmentation strategies module **204** of segmentation analyzer **110**, according to an embodiment. Segmentation strategies method **300** comprises one or more activities, which although described in a particular order may be performed in one or more permutations, according to particular needs.

[0063] After determining the number and type of supply chains, as described in connection with value offering module **200** and mapping customer clusters, product clusters, customer-product clusters, and/or customer business models to supply chain models, as described in connection with clustering module **202**, segmentation strategies module **204** may perform segmentation strategies method **300** to provide an end-to-end approach to configure and directly reflect different service priorities in different supply chain planning modules differentiating both the solution design and configuration of each module. According to embodiments, segmentation strategies method **300** determines configurations that will support a supply chain process that can handle the different characteristics required to support the supply chain models defined by the value offering module **200** and associated with product-customer associations by the clustering module **202**. Additionally, or in the alternative, segmentation strategies module **204** automatically configures the supply chain management planning and execution modules to make the supply chain “Agile,” “Collaborative,” “Efficient Collaborative,” “Productive,” or “Responsive” for particular products and customers, according to the customer hierarchy and prioritization grid generated by clustering module **202**.

[0064] At action **302**, segmentation strategies module **204** defines steering options. At action **304**, segmentation strategies module analyzes customer hierarchy to determine forecast and allocation level for customers, products, and customer-product combinations. At action **306**, segmentation strategies module **204** designs strategies, levers, and configuration options. At action **308**,

segmentation strategies module **204** differentiates configuration options for each by supply chain model. Segmentation strategies module **204** defines which configuration options will differentiate the service and provide differentiated costs to each of these customer-product combinations. [0065] At action **310**, segmentation strategies module **204** implements the configuration options across the supply chain network. For example, based on the supply chain models, the supply chain may be altered from a single static supply chain to more than one active supply chains that require, for example, changing the hierarchy and priority of customers and products or the transportation of the products. In addition, embodiments of segmentation strategies module **204** may alter one or more modules of supply chain planning and execution processes, end-to-end, beginning with, for example, demand planning and ending with, for example, order promising. This may include, for example, altering and configuring all planning processes and execution processes differently, including implementing levers **504** (FIG. 5) for each supply chain model.

[0066] Each of the actions of the clustering method **300** are explained in more detail below.

1. Defining Steering Options

[0067] At action **302**, segmentation strategies module **204** may define steering options. According to embodiments, steering options are defined based, at least in part, on one or more performance radars, which may be determined by value offering module **200** and derived from one or more supply chain models. Steering options determine how service priorities may be achieved across the various supply chain planning and execution modules.

[0068] FIG. 4 illustrates a simplified segmented supply chain network **400** with various exemplary supply chain models and various exemplary steering options, according to an embodiment. Steering options may comprise differences in supply chain processes or services that result in different outcomes to one or more customers. For example, for exemplary customers such as retailer **158** and a wholesaler, retailer **158** and wholesaler may require different services or outcomes from the supply chain. To create that outcome, the supply chain may be configured with different steering options to alter the behavior the supply chain, such as, for example, using different levers to alter one or more supply chain planning and execution processes.

[0069] As illustrated in the simplified segmented supply chain network **400**, a manufacturing plant **402** of manufacturer **154** may be associated with a plant warehouse (PWH), which indicates a dedicated warehouse associated with manufacturing plant **402**, and/or European Distribution Center (EDC) **404**. Manufacturer **154** may store large amounts of inventory in the PWH/EDC **404**. This inventory may be used by customers associated with the strategic supply chain model. Several Regional Distribution Centers (RDC) **406a-406c** may receive replenishment of inventory from the PWH/EDC **404**. The inventory may be replenished in different amounts, as represented by the differing number of boxes next to each RDC **406a-406c**. Replenishment is indicated by solid line **410**. Four exemplary customers **430-436** receive inventory from the RDC **406a-406c** or directly from PWH/EDC **404** as indicated by dashed line **420**. Each of these customers **430-436** is associated with a different supply chain model, which determines how the customer will be supplied with inventory from different locations in segmented supply chain network **400**. To further explain how supply chain models determine customer service levels, several examples are now given.

[0070] Each supply chain model comprises key characteristics that determine a particular service offering for customers associated with that supply chain model. Five exemplary supply chain models are illustrated: efficient, responsive, agile, collaborative, and strategic. Each delivers a different service package for specific types of customers and products. An efficient supply chain model may be associated with cost effective service packages (such as, for example, Make-To-Order (MTO)). A responsive supply chain model may be associated with a standard service package (such as, for example, a default service level). An agile supply chain model may be associated with a premium service package. A collaborative supply chain model may be associated with a customer collaboration service package. A strategic supply chain model may be associated

with a regional importance service package. Each of the supply chain models may be configured with different steering options to enforce the outcome desired for each customer and product. Each steering option represents a lever **504** (FIG. 5) which, when configured, alters one or more supply chain planning and execution processes according to the configuration options **508** associated with each particular lever **504**.

[0071] For example, as illustrated in the above figure, a steering option may comprise, for example, forecast level lever **442**, demand prioritization lever **444**, stocking strategy lever **446**, production rules lever **448**, and ATP allocation lever **450**.

[0072] Forecast level lever **442** may comprise forecasting demand and supply at different hierarchical levels, as described in more detail below. Demand prioritization lever **444** may comprise prioritizing the demand differently for different customer groups **444**. Stocking strategy lever **446** may comprise defining a different stocking strategy whereby for some groups the inventory is kept closer to the customer or closer to the plan in the available distribution network.

[0073] Production rules lever **448** may comprise one or more production rules such as, for example, made-to-order (MTO), engineered-to-order, made-to-stock, and the like. According to embodiments, ATP allocation lever **450** comprises forecasting a customer level and then allocating inventory at the same level.

[0074] According to some embodiments, levers **504** comprise one or more module configuration rules **508**, which create different behaviors and different outcomes for specific customer groups, as described in more detail below.

[0075] By way of a first example, a collaborative supply chain model has a characteristic of regular data exchange with customers **430**. This may mean that customers **430** share a demand forecast, replenishment data, and inventory levels each day with the supplying company. Sharing the demand forecast, replenishment data, and inventory levels with the company allows the company to collaborate with customer **430** to ensure that the company's production and shipping plans will meet the demand forecast and inventory levels for customers **430** associated with the collaborative supply chain model at a lower cost and with less safety stock. Based at least in part on the increased visibility, the company may anticipate replenishments, deliveries, or even production to meet sudden changes in the demand forecasts and inventory levels of customers **430**.

[0076] By way of a second example, an efficient supply chain model may be associated with a made-to-order business. For these customers **436**, manufacturer **154** may ship large quantities directly from a manufacturing plant **402** to the made-to-order business without storing any inventory in the segmented supply chain network **400**, leading to significant cost savings in production and storage.

[0077] By way of a third example, customers **432-434** associated with an agile supply chain model and responsive supply chain model may receive inventory from one or more RDCs **406b-406c**. For customer **432** associated with the agile supply chain model, orders are placed in an ad-hoc pattern which requires RDC **406b** to keep a large inventory closer to customers **432** and only use PWH/EDC **404** if order lead-times allow such a delay. In contrast, for the responsive supply chain model, customers **434** order at regular intervals in large quantities which allows RDC **406c** to keep a lower inventory and instead provides for regular and frequent full truckload plant shipments directly from PWH/EDC **404**.

[0078] FIG. 5 illustrates differentiator hierarchy **500** for module configuration, according to an embodiment. Module configuration may require one or more differentiators. A high-level differentiator may comprise various supply chain strategies **502**. A mid-level differentiator may comprise levers **504**. A low-level differentiator may comprise enablers **506**. At a general level, each supply chain strategy **502** relates to a supply chain model (such as agile, responsive, collaborative, and the like.) To cause the supply chain to reflect the chosen supply chain strategy **502**, one or more levers **504** and one or more enablers **506** are activated to support the chosen supply chain strategy **502**. Each supply chain strategy **502** may be associated with various levers **504** to modify

functions of the modules to behave differently. For levers **504** to operate in the modules, one or more enablers **506** may need to be activated.

[0079] Levers **504** comprise primary differentiators between supply chain strategies. Enablers **506** comprise additional functions that may be needed to activate supply chain strategy **502**. Each lever **504** and enabler **506** may be associated with one or more configuration options **508a-508f**.

Configuration options **508a-508f** comprise a particular implementation of one or more implementations of levers **504** and enablers **506** that differentiate that lever **504** or enabler **506** from others. To further illustrate supply chain strategies **502**, levers **504**, and enablers **506**, an example is now given.

[0080] FIG. **6** illustrates an exemplary overview **600** of supply chain planning and execution processes with various exemplary supply chain strategies **502** and levers **504**, according to an embodiment. According to embodiments, a supply chain comprises various modules for planning and execution processes which interact to perform various supply chain operations, including demand planning, inventory planning, production planning, and the like. According to embodiments, segmentation strategies module **204** modifies the various supply chain planning and execution processes using supply chain strategies **502** to enforce one or more selected supply chain models and altering the supply chain operations. As noted by various processes, M, W, and D indicate whether a process is performed in the exemplary supply chain on a monthly (M), weekly (W), or daily (D) basis. Monthly processes may comprise longer-term planning processes. Weekly processes may comprise update processes (which may occur each weekend, for example). Daily processes may comprise transactional processes, such as, for example, making orders on a day-to-day basis or shipping inventory on a day-to-day basis.

[0081] According to embodiments, demand planning process **602** may comprise a process comprising forecasting demand by one or more companies or customers. These forecasts **604a-604c** may then be transmitted for use to one or more other processes for determining the appropriate inventory levels, such as, for example, to inventory planning process **606**, tactical master planning process (MP-TA) **608**, and/or operational master planning process (MP-OP) **610**. in order to balance demand and supply for both tactical and operational supply planning horizons. According to embodiments, tactical master planning process **608** and operational master planning process **610** balance demand and supply over these two distinct horizons, supported by the evaluation of various potential demand and supply scenarios using demand and supply balancing (DSB), which analyzes and evaluates various potential business scenarios. As part of sales and operations planning (S&OP), demand and sales balancing may occur at monthly frequency to evaluate gaps between forecasted demand and supply plans, which may be constrained due to, for example, missing production capacity or raw material availability. Although demand and sales balancing is described as occurring monthly, demand and sales balancing may occur at other frequencies, such as daily, weekly, or the like. According to embodiments, tactical master planning process **608** using demand and supply balancing may generate a DSB decision **612** comprising, for example, market and product priorities such as an amount of production to short, solutions to minimize lateness, and/or how to allocate scarce production capacity to meet market demands. For instance, continuing with the example discussed above of an exemplary tire manufacturer, if demand planning process **602** projects a demand forecast **604a-604c** of 10 million tires of different types, the tire company may then determine production capacity and balance the available supply, which may be more or less than 10 million tires, with the projected demand. According to embodiments, inventory planning process **606** generates inventory norms **614**.

[0082] After balancing demand and supply, operational master planning process **610** may create a feasible supply chain master plan, which may then be translated by different supply chain processes or activities, including, for example, raw material planning process **616**, production planning process **618**, and replenishment planning process **620**. Raw material planning process **616** may use the master plan to procure raw materials for production (such as raw materials to produce tires for

the exemplary tire manufacturer). Production planning process **618** may use the master plan to create the details of a production plan along one or more production lines. Replenishment planning process **620** may use master plan to drive distribution activities, such as, for example, the transportation of items between different nodes in supply chain network **100**.

[0083] Supply plan **622** is based, at least in part, on master planning (MP-TA, MP-OP) decisions (both tactical and operational) about, for example, production priorities, shorting production for certain products and customers, and the like, as a result of the demand=supply balancing, described above. According to embodiments, supply plan **622** reflects business priorities that meet the demand with the available production capacity. Generating supply plan **622** provides for aligning production plans with business, customer and product service priorities, leveraging available capacity to meet business targets, and producing according to agreed business and market priorities.

[0084] Customer allocation process **624** comprises allocating available or future planned inventory for future incoming orders. Based on one or more allocation rules, the inventory for each product, resulting from supply plan **622**, and either stored in a warehouse or on its way to the warehouse (such as, for example, in planning, in production, in transit, and the like) is allocated to various customer groups, based, at least in part, on one or more predefined rules (such as, for example, as per forecast, as per split %, and the like) According to embodiments, customer allocation process **624** makes available inventory virtually available to different levels of customers with different service priorities (reservations) or kept free for all customers to consume. Customer allocation process **624** comprises a reservation system for current and future available inventory. Depending on the customer, orders will be able to consume either all available inventory or only a small and allocated part of the inventory. By leveraging this concept, manufacturers **154** and retailers **158** may prevent first-come-first-serve inventory consumption, where all inventory is freely available for everybody. For example, customer allocation process **624** may allocate certain parts of an inventory to higher priority customers, and protect it from orders from other customers.

[0085] Allocation plan **626** may comprise at least two types of supply for customers. Available-to-Promise (ATP) comprises an original supply plan **622** consisting of current and future available inventory. ATP may comprise, for example, an available supply of items for fulfilling orders, such as, for example, items stored in one or more warehouses, distribution centers **156**, or the like. Some portion of ATP inventory may be transformed into allocated-available to promise (AATP) and is not available for ordinary customers. One or more allocation rules specify which customers may consume inventory allocated to each type of customer. The one or more allocation rules may reserve inventory and may be company specific. Allocation plan **626** comprises the calculated reserved and free inventory showing ATP (e.g. available supply) and how it was converted into AATP (e.g. reservations).

[0086] Order promising process **628** comprises checking in real time, as orders received by an order entry system and/or segmentation analyzer **110**, against inventory projections, ATP, and AATP, to reserve inventory to meet each order and calculate a promise date for each order. As described in more detail below, some customers' orders will be allowed to consume only free inventory. Other customers' order will consume from AATP, which comprises guaranteed inventory and may not be consumed earlier orders of other customers.

[0087] According to embodiments, order promising process **628** is performed by an AP-OP engine and comprises a first action, for calculating AATP based, at least in part, on available inventory from supply plan **622**, a second action, for checking incoming orders against AATP, and a third action, for calculating a reliable due date or order promise, based, at least in part, on reserved inventory. Reserved inventory comprises, for example, one or more available items, such as, for example, one or more available warehouse items or future available items based on supply plan **622**. According to embodiments, the due date is calculated based, at least in part, on for example, a leadtime to a customer from a local warehouse or RDC, whether a direct shipment is allowed for

that customer, and the like. Additionally, or in the alternative, the order entry system receives confirmed orders with a reliable due date promised by the AP-OP engine and, generally, happens in real time, wherein, for each order, AP-OP engine scans AATP to provide a reliable, fast, and accurate order promise confirmation with the correct due date.

[0088] Confirmed orders **630** comprises orders received from the order entry system by the AP-OP engine and for which a reliable due date has been promised. Customer orders **632** comprise orders from an order entry system and are electronically communicated to demand planning process **602** and/or order promising process **628**. Customer orders **632** may be used to create a demand plan comprising netted forecasts and orders. Additionally, or in the alternative, customer order **632** are received by order promising process **628** and given a reliable promised due date and confirmation.

[0089] Order handling and delivery process **634** comprises releasing an order for execution after, for example, a credit check, and setting up a transport order to ship the customers' orders, after a reliable due date is received. Delivery orders **636** comprise executed delivery orders, passed to transportation network **140**, to ship products to the right customers, with any necessary logistics and commercial documentation. Logistics and transport process **638** comprises execution of delivery orders **636**. Shipment plan **640** comprises a sum of all shipments to execute all delivery orders created in an order entry system and ready to be shipped by transportation network **140**. [0090] To avoid double counting of demand, the orders received by the order entry system, may be subtracted from the forecasted demand twice in this end-to-end process. A first netting may be performed prior to the creation of supply plan **622**. According to embodiments, a forecast is converted into a net forecast by subtracting existing orders. A second netting may be performed prior to replenishment planning **620** as the total forecast must be reduced by the orders that already are entered in the order entry system. Both for supply planning and replenishment planning **620**, the total demand to plan is the net forecast plus the existing orders (subtracted from the total forecast).

[0091] To further describe the demand fulfillment process, an example is now given of replenishment plan being translated into logistic activities. For example, after the number of inventories moved between different nodes is determined, transportation is determined by one or more transport companies in transportation network **140**. After determining how much inventory is being transported, demand fulfillment processes may be determined to allocate inventory to different customer groups. For the exemplary tire manufacturer, this may mean keeping 50% of inventory in particular warehouses for car manufacturers, keeping 30% for wholesalers, and 20% for retailers.

[0092] Once the allocation is done, then promises may be made against the allocation. For example, if a first warehouse, Warehouse A, stores 50,000 tires available for a customer group, Customer Group B, when Customer Group B enters an order for tires through the company's order system, the company may adjust the inventory and promise these orders against the available inventory in real time. The company will know how much inventory is in the warehouse, the size of the orders, and so may check each order against the available inventory and promise in real time, including the kind of available inventory and the time it will be available, for each customer that placed an order. According to embodiments, this may comprise Available Allocated to Promise (AATP), which is a function for companies to give a reliable due date for each order for each customer.

[0093] As discussed above, segmentation strategies module **204** configures supply chain modules and processes (including any supply chain management software which operates the modules of planning and execution processes, which may be operated on or in communication with one or more servers or databases of segmentation analyzer **110**, inventory system **120**, one or more imagers **130**, transportation network **140**, one or more supply chain entities **150**, and/or computer **160** of supply chain network **100**) to enforce priorities determined by the supply chain models, and, in some cases, may override supply chain actions to enforce the supply chain models. In the

exemplary overview **600**, segmentation strategies module **204** has configured the supply chain modules and processes using an active inventory management supply chain strategy **650** (associated with inventory planning process **606**), active demand management supply chain strategy **652** (associated with demand planning process **602**), active supply management supply chain strategy **654** (associated with operational master planning process **610**), active production management supply chain strategy **656** (associated with raw material planning process **616**), active AATP management supply chain strategy **658** (associated with customer allocation process **624**), and active shipment management supply chain strategy **660** (associated with order handling and delivery process **634**). Although particular segmentation strategies are associated with particular supply chain processes, embodiments contemplate any number of segmentation strategies associated with various supply chain processes, according to particular needs.

[0094] For example, segmentation strategies module **204** may configure order and handling delivery process **634** with active shipment management supply chain strategy **660** to override a supply chain action that would ship a product to a low priority customer before a high priority customer. As discussed below, a customer associated with a collaborative supply chain model may have higher priority than a customer associated with a responsive supply chain model. According to embodiments, segmentation strategies module **204** configures the supply chain management software to prioritize, for example, product orders for the customers associated with the collaborative supply chain model (who may be paying a higher cost to the company for the prioritized service). According to another example, segmentation strategies module **204** may configure the supply chain management software to hold inventory for the customers associated with a higher priority supply chain model, even when an order has been placed by a customer associated with a lower priority supply chain model, by using an active AATP management supply chain strategy **658**.

[0095] Additionally, in the exemplary overview **600**, segmentation strategies module **204** has configured the supply chain modules and processes using various levers **504**, including, opportunity demand lever **670** (associated with tactical master planning process **608**), common size lever **672** (associated with tactical master planning process **608**), and steering across ICOs lever **674** (associated with operational master planning process **610**). Although particular levers **504** are associated with particular supply chain processes, embodiments contemplate any number of levers **504** associated with various supply chain processes, according to particular needs.

[0096] As discussed above, supply chain strategies **502**, levers **504**, and enablers **506** are based, at least in part, on a customer hierarchy generated by clustering module **202**.

[0097] FIG. 7 illustrates an exemplary customer hierarchy **700**, according to an embodiment. At action **304**, segmentation strategies module analyzes customer hierarchy to determine forecast and allocation level for customers, products, and customer-product combinations.

[0098] For example, segmentation strategies module **204** determines at which level **702** of the hierarchy forecasts and allocations of products are made for each customer in customer hierarchy **700**. Forecasts and allocations of products may be made at different levels of the hierarchy depending on the supply chain model associated with the customers in customer hierarchy **700**. Forecasts and allocations may be determined for an individual customer, a group of customers, or a region (such as a country), or any other grouping or level of hierarchy.

[0099] Customer hierarchy **700** defines the level **702** at which planners manage forecasts and allocations of the supply chain. According to an embodiment, forecasting and allocation may be done at different levels **702** in hierarchy **700**, depending on cluster **704**. Levels **702** comprise a country level (which may also be referred to as an intercompany entity (ICO) level **710** (e.g. a market level)), a steering level **712**, a cluster level **714**, and a planning customer code number (planning CCN) level **716**. According to embodiments, clustering module **202** adjusts customer hierarchy **700** by defining an intermediate level of granularity that balances between forecasting and allocating all products for each customer and forecasting and allocating all products at a

country or regional level for these customers.

[0100] As illustrated, ICO level **710** contains only the ICO. Steering level **712** comprises central steering **706** (the regional organization controlling regional inventory allocation decisions) and market steering **708** (the country organizations controlling more local inventory allocation decisions in each country). Cluster level **714** comprises the supply chain models associated with each customer cluster (i.e. Strategic, Collaborative, Agile, and Responsive, and the like). Clusters **704** may each be associated with a supply chain model such as, for example, Cluster 1 **720** may be assigned to a strategic supply chain model, Cluster 2 **722** may be assigned to a collaborative supply chain model, Cluster 3 **724** may be assigned to an agile supply chain model, and Cluster 4 **726** may be assigned to a responsive supply chain model.

[0101] Planning CCN level **716** comprises the various CCNs (i.e. CCN11, and various CCN2s). Additionally, the CCN2 of an agile supply chain model may comprises two “sold to” boxes: sold to 1 and sold to 2. In this instance, allocation of inventory may be set to even a finer granularity than customer granularity, such as, for example, allowing the local market organizations to allocate inventory, because, for some customers it is critical to guarantee availability and service to a particular ship-to location, to avoid the other ship-to location for that same customer orders all the tires and consumed the available inventory.

[0102] For example, allocation comprises a process of taking a committed supply plan **622** and defining how available future inventory for different products will be made available to different customer segments. By way of example and not by limitation, if a million tires will become available in regional distribution center **406a-406c** in a particular region, an allocation process may comprise determining which part will be reserved (for particular customers or pools of particular customers) or be available as free inventory to all customers. An allocation process may carry out these allocations based on rules, such as per committed forecast, fixed split, or fixed quantity to make sure inventory cannot be consumed on a first-come-first-serve basis by lower priority customers, before higher priority customers placed their orders. As such, allocation may protect the inventory and the future service for the higher priority customers.

[0103] Governance indicates whether planning CCNs are steered centrally **706** or by markets **708**, as indicated at steering level **712**, and as indicated in more detail below. Continuing with the exemplary tire manufacturer, a planning CCN provides each single market to allocate market open tires in a controlled away. Instead of forecasting each customer, markets may forecast groups of ‘same service’ customers into a predefined grouping of customers, planning CCNs, and allocate or reserve a specific tire quantity for them.

[0104] For example, continuing with the example of a tire manufacturer, segmentation strategies module **204** may determine that a forecast will be made at a customer level for a big-box retailer with stores nation-wide that sells tires from the tire manufacturer. The forecast may be made at customer level for the big-box retailer based on the big-box retailer being associated with a collaborative supply chain model. The collaborative supply chain model may have been associated with the big-box retailer because the big-box retailer is a strategic customer of the company that the big-box retailer has a service level agreement with the company, or other considerations that determine the supply chain model associated with the big-box retailer, as explained below. On the other hand, a small retailer, such as a mom-and-pop store may be forecasted with all other mom-and-pop stores at the country level because they are associated with a responsive supply chain model. The responsive supply chain model may have been associated with the mom-and-pop stores because they are too small or too numerous to generate individual forecasts, they are not a strategic customer of the company, they do not have a service level agreement, or other like considerations.

[0105] Similar to the determination of the forecasting level, segmentation strategies module **204** determines appropriate levels of detail for allocations of inventory based on similar considerations.

[0106] FIG. **8** illustrates exemplary allocation logic **800** for various supply chain models associated with customer hierarchy **700**, according to an embodiment. According to the exemplary allocation

logic **800**, customer clusters **802-808** are associated with four exemplary supply chain models, “Strategic,” “Collaborative,” “Agile,” and “Responsive.” A responsive supply chain model may be forecast at customer cluster level **714**, an agile supply chain model may be forecast separately from the agile customers and at planning CCN level **716**. In other words, wholesale-like customer cluster **808** is associated with a responsive supply chain model and these customers may be forecast in connection with Cluster 4, while retail-like customer cluster **806** are associated with an agile supply chain model and are forecast in connection with Cluster 3. Collaborative customers cluster **804** are forecast with Cluster 2 and Strategic customer cluster **802** are forecast with Cluster 1. According to some embodiments, allocation and forecasting are done for the same clusters and levels. Therefore, a strategic customer in strategic customer cluster **802** would receive priority and first allocation of tires, so they are associated with Cluster 1.

[0107] Additionally, based on the particular cluster assignment, customers may receive products that have been promised to inventory for customers in a lower cluster assignment. For example, if a customer associated with a strategic supply chain model orders more product than forecasted, the company may deliver product that was promised to inventory for customers in a lower cluster. On the other hand, a customer associated with a collaborative supply chain model would be able to take inventory promised to customers associated with Cluster 3 and Cluster 4, but not from Cluster 1. According to embodiments, the cluster assignment may automatically enforce a prioritization that has been assigned to the customers.

[0108] By way of a more particular example, a big box retailer may be assigned a strategic supply chain model based on a service level agreement, whereas a mom-and-pop store might be assigned to a responsive supply chain model. The big box retailer may have reserved inventory, stored at one or more strategic inventory locations, whereas the mom-and-pop store receives only free inventory. The big box retailer will receive first priority to inventory, whereas the mom-and-pop store will only receive inventory after all other customers' needs have been met. Customers assigned to one or more intermediate priorities between strategic customers and responsive customers may receive pooled inventory.

[0109] To further explain allocation logic **800**, another example is now given. As discussed above, an allocation of products might not be identical for all products. For example, super strategic products may represent, for example, specialized tires for luxury cars, produced in small batches with absolute top priority. These super-strategic tires would be managed and allocated to various customers centrally. Regional strategic tires, may comprise, for example, special ‘private label’ tires, with the brand of the distributor distributed across the region. Market specific tires may comprise, for example, a local brand very popular in a particular country with an exclusive partnership agreement between the tire manufacturer and the local national distributor. Market open may comprise products that are provided in a fair-share manner to any customer in a market. Each market would have the autonomy about how to allocate the available inventory to its own customer base.

[0110] The above overview of an exemplary supply chain process services many different customer groups. In the above methods, the company has defined different support options for each of the customers. Therefore, along these processes, the company needs to differentiate the services to provide the outcomes chosen by the supply chain models.

2. Design Strategies, Levers, and Configuration Options

[0111] At action **306** of segmentation strategies method **300**, segmentation strategies module **204** may design strategies **502**, levers **504**, enablers **506**, and configuration options **508**.

[0112] To provide differential service, segmentation strategies module **204** may define one or more supply chain strategies **502** to enable the supply chain model. As illustrated in FIG. 6, segmentation strategies module **204** selected supply chain strategies **502** comprising active inventory management supply chain strategy **650**, active demand management supply chain strategy **652**, active supply management supply chain strategy **654**, active production management supply chain

strategy **656**, active AATP management supply chain strategy **658**, and active shipment management supply chain strategy **660**. Although these particular supply chain strategies **502** are chosen for this supply chain, embodiments contemplate any suitable number or combination of supply chain strategies **502**, according to particular needs.

[0113] FIG. **9** illustrates supply chain strategies **502** selected to differentiate supply chain models, according to an embodiment. According to embodiments, each supply chain strategy **502** comprises a default strategy **902** and special case strategies **904**. For example, an active demand management supply chain strategy **652** may comprise setting the detail level at which forecasts are made. By default, a company may forecast by market, such as by region, state, country, or the like. For a responsive supply chain, that level of granularity is adequate. However, for a customer that has a service level agreement, such as a big box retailer, the customer may need a higher level of service. This may require a more granular forecast and to forecast that customer separately from other customers in the same market. In other words, a responsive supply chain may forecast for the market level, but for higher service levels (such as an agile or collaborative supply chain model), the level of granularity is higher or finer.

[0114] By way of a further example, active AATP management supply chain strategy **658** may be improved by a more granular forecast. Active AATP management supply chain strategy **654** provides an advantage of the company working directly with the customer to make the quality of the forecast better. Because the forecast is more accurate, the company may improve allocation as well. The company may, for example, keep the inventory for a particular customer separate from other customers, so that the company may keep AATP or a virtual protected inventory in a warehouse specifically for the customer.

[0115] Active supply management supply chain strategy **654** comprises prioritizing supply. If a capacity for production is not sufficient, some items will be produced late or not produced at all. According to embodiments, active supply management supply chain strategy **654** configures one or more supply chain planning and execution processes to prioritize and choose what items will be delayed, such as being produced or supplied late, or choosing what customers will be supplied late or have their items produced late. In other words, active supply management supply chain strategy **654** may choose which customers' orders are prioritized over the orders of other customers by allocating capacity to some customers before others. For the exemplary tire manufacturer, this may comprise prioritizing car manufacturers over, for example, mom-and-pop stores or stores that offer replacement tires.

[0116] Active inventory management supply chain strategy **650** may comprise allocating where inventory is stored, such as, for example, in a plant warehouse, regional distribution center, customer warehouse, or the like. Depending on the supply chain model, one or more supply chain planning and execution processes will position inventory at different places in supply chain network **100**, with a goal to keep the inventory as far away from the customer as possible (in order to give more flexibility of deploying the inventory), while still meeting all the needs of the customers. For example, if an entire order is stored in Tennessee, and an order is received from New York, the company will face additional costs and delays to move the tires between the two regions. Therefore, it is more cost effective to keep the tires at the manufacturing plant and move them once as the order is received. However, for some customers, tires are expected at the store within four hours. Keeping the tires at the plant warehouse would prevent the company from meeting this timeline. Therefore, the inventory for customers with short order times must be kept at storage facilities closer to the individual stores for that customer.

[0117] Active shipment management supply chain strategy **660** may comprise determining the proper allocation of inventory to one or transportation vehicles **146**. There is always a tradeoff between completely filling a transportation vehicle **146**, such as a truck, and having enough order to fill the transportation vehicle **146** before it leaves a storage facility. For some supply chain models, transportation network **140** may determine that transportation vehicle **146** must be

completely full; for other supply chain models, transportation vehicle **146** may be partially full, or to even provide for parcel service.

[0118] Active production management supply chain strategy **656** determines advance production techniques for products based on an associated supply chain model. According to embodiments, some products may be made as they are needed and other products may need to be made in large supplies in advance. For example, a company, such as the exemplary tire manufacturer, may supply tires to a customer, such as a large car manufacturer. Based on the contracts between the company and the customer, if the company fails to deliver tires, the company may face penalties that could exceed millions of dollars per day. In that example, active production management supply chain strategy **656** may ensure that the production is always sufficient to supply the forecasted needs for the customer. On the other hands, some customers may need a private label order, which may be a large value tire at a low price. In this case, active production management supply chain strategy **656** may wait for the order to begin production without building any stock in advance. In some cases, made-to-stock is a default production strategy, made-to-order is used for commodity and private label orders.

[0119] FIG. **10** illustrates supply chain levers **504** associated with various supply chain strategies **504** and configuration options **508**, according to an embodiment. As discussed above, levers **504** comprise configurable features in various modules for supply chain planning and execution processes. For example, a forecast lever **442** provides for forecasting at different levels of the supply chain, such as, for example, ICO level **710** (e.g. a market level), a country level, state level, customer level (e.g. Planning CCN level **716**), cluster level **714** (or a group of customers), or any other like level, according to particular needs.

[0120] According to a further example, active inventory management supply chain strategy **650** may be effected by stocking strategy lever **446**. The stocking strategy may be chosen between, for example, upstream-high, downstream-low, upstream-low, downstream-high, or none. This indicates, for example, that for MTO customers, the company will not store inventory. In supply chain models that postpone, inventory is kept at a high level, for example, upstream-high. And, for supply chain models where the inventory is kept close to the customers, downstream-high may be chosen.

[0121] For active supply management supply chain strategy **654**, an associated lever **504** may comprise demand prioritization lever **444**. A default for demand prioritization lever **444** may be fair share, which treats all demands equally. However, demand prioritization lever **444** may be configured to prioritize certain customers, products, or clusters of customers and products.

[0122] As indicated above, customers, products, and customer-product clusters each may comprise an associated priority. In situations where production capacity is not enough to meet a forecasted volume, the last customer, products, or customer-product clusters that will be shorted will be the ones with the highest priorities. In other words, supply chain production planning process **618** will short the demand for the customers and products that are associated with lower priority clusters. The production planning process **618** may determine priority based on a ranking of customers, products, or clusters, according to particular embodiments.

[0123] Active production management supply chain strategy **656** may comprise production rule lever **448**. According to embodiments, production rule lever **448** comprises a made-to-stock default production rule and an exception made-to-order production rule for, for example, private label customers, as explained in detail above.

[0124] Active AATP management supply chain strategy **658** may comprise ATP allocation lever **450**. Configuration options **508** for ATP allocation lever **450** may be similar to configuration options **508** for forecast level lever **442**. ATP allocation lever **450** may, for example, forecast a customer level and ATP allocation lever **450** may then allocate inventory at the same level. If forecasting is done at a market level, for example, then allocation may be done at the market level as well. In other words, if the forecasting is performed for a big box retailer at the customer level,

then the allocation of inventory for the big box retailer may also be performed at the customer level. If the forecast is done for an entire market, such as, for example, the entire German market, then the allocation of inventory will be stored for the entire market.

[0125] Active shipment management supply chain strategy **660** may be associated with delivery strategy lever **1004**. Delivery strategy lever **1004** may comprise, for example, a “direct ship full truck load” (DS-F TL), “direct ship lessen truck load” (DS-LTL), RDC, and expedite configuration options **508**. These configuration options **408** for various shipping strategies may be chosen to effect the supply chain model chosen for the particular customer, product, or customer-product cluster.

[0126] As mentioned above, particular configuration options **508** are chosen for each of levers **504** to alter various supply chain planning and execution processes to enforce the particular supply chain model chosen.

3. Differentiate Configuration Options by Supply Chain Model

[0127] At action **308** of segmentation strategies method **300**, segmentation strategies module **204** may differentiate configuration options **508** by an associated supply chain model for each lever **504** and supply chain strategy **502**.

[0128] FIGS. **11A** and **11B** illustrate particular configuration options **508** for various exemplary supply chain models **1102-1110**, according to an embodiment. For the five exemplary supply chain models **1102-1110** mentioned in the above example—strategic supply chain model **1102**, collaborative supply chain model **1104**, agile supply chain model **1106**, responsive supply chain model **1108**, and efficient supply chain model **1110**—each of supply chain strategies **502** requires particular configuration options **508** for each of levers **504**. Some of the configuration options **508** are the same for supply chain models **1102-1110**, and some are changed. For a default supply chain model, here, the responsive supply chain model **1108**, configuration options **508** represent default configuration options **508**. Although responsive supply chain model **1107** is a default in the above example, any supply chain model **1102-1110** may be a default, according to particular needs. In order to move customers or products to different supply chains, configuration options **508** are modified from the defaults. Exemplary changes may be made to configuration options **508** to move customers and products to different supply chains, such as agile supply chain model **1106**, efficient supply chain model **1110**, or the like, and additionally, which owner (such as central **706** (e.g. a planning department) or market **708**) configures that supply chain model.

[0129] According to some embodiments, most or all customer, products, or customer-product clusters are initially placed in a default supply chain model. To improve service to customers, various levers **504** may be configured to move customer, products, or customer-product clusters from default supply chain model to a different supply chain model, although this may increase the cost. Additionally, to improve cost, some customer, products, or customer-product clusters, various levers **504** may be configured that reduce service, but also reduce cost to customers.

[0130] FIG. **12** illustrates configuration options **508** for levers **504** for a strategic supply chain model **1102**, according to an embodiment. As discussed above, each supply chain model **1102-1110** may receive configuration options **508** that configure levers **504** that alter supply chain planning and execution process modules to change services or cost for particular customers, products, or customer-product clusters. Additionally, also as discussed above, configuration options **508** may determine at which hierarchy level **702** forecasting and allocation may be chosen, which affects service and cost by working with various levers **504** to determine, for example, where inventory is stored, how products are produced, and other like supply chain processes. Additionally, levers **504** may be inserted between particular supply chain planning and execution processes or between particular supply chain entities **150**. To configure the supply chain network **100** to enforce strategic supply chain model **1102**, configuration options **508** for forecast lever **442** are set to cluster level **714** or planning CCN level **716**, allocation lever **450** is set to cluster level **714**, demand prioritization lever **444** is set to high cluster priority, stocking strategy lever **446** is set to inherited

from previous supply chain model, production rule lever **448** is set to MTS, and delivery strategy lever **1004** is set to inherited from previous supply chain model.

[0131] FIG. **13** illustrates configuration options **508** for levers **504** for a collaborative supply chain model **1104**, according to an embodiment. To configure the supply chain network **100** to enforce a collaborative supply chain model **1104**, configuration options **508** for forecast lever **442** and allocation lever **450** are both set to planning CCN level **716** (e.g. customer level), demand prioritization lever **444** is set to ICO fair share, stocking strategy lever **446** is set to the strategy indicated in a service level agreement (SLA), production rule lever **448** is set to MTS, and delivery strategy lever **1004** is set to the strategy indicated in the SLA.

[0132] FIG. **14** illustrates configuration options **508** for levers **504** for an agile supply chain model **1106**, according to an embodiment. To configure the supply chain network **100** to enforce an agile supply chain model **1106**, configuration options **508** for forecast lever **442** and allocation lever **450** are both set to cluster level **714**, demand prioritization lever **444** is set to ICO fair share, stocking strategy lever **446** is set to upstream-low and downstream-high, production rule lever **448** is set to MTS, and delivery strategy lever **1004** is set to RDC/expedite.

[0133] FIG. **15** illustrates configuration options **508** for levers **504** of a responsive supply chain model **1108**, according to an embodiment. To configure the supply chain network **100** to enforce a responsive supply chain model **1108**, configuration options **508** for forecast lever **442** and allocation lever **450** are both set to ICO level **710** (e.g. a market level), demand prioritization lever **444** is set to ICO fair share, stocking strategy lever **446** is set to upstream-high and downstream-low, production rule lever **448** is set to MTS, and delivery strategy lever **1004** is set to DS-FTL/DS-LTL/RDC.

[0134] FIG. **16** illustrates configuration options **508** for levers **504** for an efficient supply chain model **1110**, according to an embodiment. To configure the supply chain network **100** to enforce an efficient supply chain model **1110**, configuration options **508** for forecast lever **442** and allocation lever **450** are both set to ICO level **710** (e.g. market level), demand prioritization lever **444** is set to ICO fair share, stocking strategy lever **446** is set to upstream, production rule lever **448** is set to MTO, and delivery strategy lever **1004** is set to DS-FTL.

[0135] As indicated in the above supply chain models **1102-1110**, forecasting, allocation, and configuration options all work together to determine the services provided by supply chain network **100** for particular customers, products, or customer-product clusters.

[0136] FIG. **17** illustrates a chart **1700** detailing the relationship between supply chain strategies **502**, levers **504**, and supply chain strategy steering (central steering **706** and market steering **708**). According to embodiments, levers **504** in supply chain network **100** may be configured at different locations of the supply chain network. For example, central steering **706** may comprise controlling various supply chain models **1102-1110** with configuration options **508** by a central planning organization, even though specific configuration of levers **504** may be left to market steering **708**, as they do not impact the prioritization concept that may be part of segmentation design. Therefore, the central organization or the market may determine services to customers and alter the prioritization and availability of inventory. According to embodiments, a salesperson in a distant market, such as a country foreign to the company's headquarters, may configure the priority of customers in the distant market when the customers place orders, given the allocated inventory from the central organization. If left to the team at the central organization, allocation to this level of detail may be inefficient and provide incorrect prioritization.

[0137] Reference in the foregoing specification to “one embodiment”, “an embodiment”, or “some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

[0138] While the exemplary embodiments have been shown and described, it will be understood

that various changes and modifications to the foregoing embodiments may become apparent to those skilled in the art without departing from the spirit and scope of the present invention.

Claims

1. A computer-implemented method for managing inventory by a computer comprising a processor and a memory, comprising: receiving one or more forecasts for determining one or more inventory levels; allocating available or future planned inventory for future incoming orders; allocating inventory for each of one or more products to one or more customer groups; allocating the inventory according to an available-to-promise approach and an allocated-available-to-promise approach, wherein each order is checked against available inventory and promised in real time; generating one or more inventory norms; applying one or more inventory policies; and shipping the inventory on a day-to-day basis.
2. The computer-implemented method of claim 1, wherein the one or more inventory policies comprises one or more of: a minimum order quantity, a maximum order quantity, a discount, a step-size order quantity, and batch quantity rules.
3. The computer-implemented method of claim 1, wherein the one or more inventory policies each describe a reorder point and target quantity.
4. The computer-implemented method of claim 1, wherein the one or more inventory norms comprise one or more of: a no-stockout probability, a fill rate and a cost.
5. The computer-implemented method of claim 1, wherein the inventory is allocated to the one or more customer groups based on one or more predefined rules.
6. The computer-implemented method of claim 1, wherein the available-to-promise approach and the allocated-available-to-promise approach reserves inventory and calculates a promise date.
7. The computer-implemented method of claim 6, wherein the reserved inventory comprises one or more available warehouse items or future available items based on a supply chain.
8. A system for managing inventory, comprising: a computer comprising a processor and a memory and configured to: receive one or more forecasts for determining one or more inventory levels; allocate available or future planned inventory for future incoming orders; allocate inventory for each of one or more products to one or more customer groups; allocate the inventory according to an available-to-promise approach and an allocated-available-to-promise approach, wherein each order is checked against available inventory and promised in real time; generate one or more inventory norms; apply one or more inventory policies; and ship the inventory on a day-to-day basis.
9. The system of claim 8, wherein the one or more inventory policies comprises one or more of: a minimum order quantity, a maximum order quantity, a discount, a step-size order quantity, and batch quantity rules.
10. The system of claim 8, wherein the one or more inventory policies each describe a reorder point and target quantity.
11. The system of claim 8, wherein the one or more inventory norms comprise one or more of: a no-stockout probability, a fill rate and a cost.
12. The system of claim 8, wherein the inventory is allocated to the one or more customer groups based on one or more predefined rules.
13. The system of claim 8, wherein the available-to-promise approach and the allocated-available-to-promise approach reserves inventory and calculates a promise date.
14. The system of claim 13, wherein the reserved inventory comprises one or more available warehouse items or future available items based on a supply chain.
15. A non-transitory computer-readable medium embodied with software, the software when executed configured for managing inventory by: receiving one or more forecasts for determining one or more inventory levels; allocating available or future planned inventory for future incoming

orders; allocating inventory for each of one or more products to one or more customer groups; allocating the inventory according to an available-to-promise approach and an allocated-available-to-promise approach, wherein each order is checked against available inventory and promised in real time; generating one or more inventory norms; applying one or more inventory policies; and shipping the inventory on a day-to-day basis.

16. The non-transitory computer-readable medium of claim 15, wherein the one or more inventory policies comprises one or more of: a minimum order quantity, a maximum order quantity, a discount, a step-size order quantity, and batch quantity rules.

17. The non-transitory computer-readable medium of claim 15, wherein the one or more inventory policies each describe a reorder point and target quantity.

18. The non-transitory computer-readable medium of claim 15, wherein the one or more inventory norms comprise one or more of: a no-stockout probability, a fill rate and a cost.

19. The non-transitory computer-readable medium of claim 18, wherein the inventory is allocated to the one or more customer groups based on one or more predefined rules.

20. The non-transitory computer-readable medium of claim 15, wherein the available-to-promise approach and the allocated-available-to-promise approach reserves inventory and calculates a promise date.
